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μ-analysis based algorithms for probabilistic finite-frequency H_2-performance analysis of uncertain linear systems

Franca Somers[1], Clément Roos[1], Valentin Preda[2] and Francesco Sanfedino[3]

***Abstract*— Current industrial validation and verification (V&V) activities mostly rely on time-consuming Monte Carlo simulations. They are able to quantify the probability of sufficiently frequent phenomena, but they may fail in detecting rare but critical combinations of uncertainties. In recent years, deterministic model-based worst-case analysis methods have reached a good level of maturity. Without the need of simulations, they explore the entire space of all possible combinations of uncertain parameters, and provide guaranteed mathematical bounds on robust stability margins and worst-case performance levels. Identified problematic parameter configurations can then be used to guide the Monte Carlo campaigns and shorten the V&V process. But a limitation of these deterministic methods is that the probability of occurrence of critical parameter combinations is not measured, which means that a control system can be rejected based on an extremely unlikely scenario. Probabilistic μ-analysis has therefore been studied since the 1990s to bridge this analysis gap between traditional simulation-based and deterministic worst-case analysis techniques, by focusing on rare but nonetheless possible situations that may threaten system integrity. In this context, this paper introduces three novel μ-based probabilistic H_2-analysis algorithms, which provide guaranteed upper and lower bounds on the probability that an uncertain system violates a desired H_2-performance requirement on a fixed frequency range.**

I. Introduction

Thanks to its multiple interpretations, the H_2 norm has become a common measure for control laws design and validation in aerospace engineering, see e.g. [1], [2]. Firstly, it captures the system response to stationary noise. In other words, it measures the variance of the system output when the input is white noise, which is for example crucial information for high-accuracy pointing space missions. Secondly, by Parseval's theorem, the H_2 norm of a linear system corresponds to the L_2 norm of its impulse response [3]. This gives insight into the transient response of the system. For example, this can be particularly useful when studying the "ringing" effect caused by small particle collisions on a spacecraft. Finally, an evident relation with the observability and controllability Gramians can be pointed out [4].

The worst-case H_2-performance problem for the study of uncertain linear systems was first introduced in the 1970s, in an effort to provide stability margins for the H_2-optimal Linear Quadratic Gaussian (LQG) controller [5]. However, the difficulties encountered when combining classical and modern control techniques led to a shift in focus towards alternative performance criteria, like the currently very mature worst-case H_∞-analysis theory. In the 1980s, there was a renewed interest in the strength of the H_2 norm. This mainly led to new techniques that combine H_2- and H_∞-analysis, but also to new ideas on pure worst-case H_2-performance analysis. The survey by Paganini and Feron [5] gives a detailed overview of the work carried out between the late 80s and early 90s, which mostly led to Linear Matrix Inequalities (LMIs) based methods in both the time and frequency domains.

Following on from these early contributions, Paganini proposed both frequency-domain and state-space conditions to compute an upper bound on the worst-case H_2 norm [6], [7]. The state-space based approach calculates an upper bound using a single LMI, but the dimension of the problem rapidly grows with the number of states. For the frequency-based approach, the number of decision variables depends only on the number of uncertainties and their occurrences in the model, but the number of LMIs is infinite. In practice, the problem is solved on a finite frequency grid, and a Hamiltonian-based technique inspired by [8] is proposed in [9] to validate the solution on the entire frequency range.

More recently, robustness analysis approaches based on Integral Quadratic Constraints (IQC) were proposed in [2] and [10]. They allow to extend the classical robustness analysis of time-invariant parametric and dynamic uncertainties to the analysis of time-varying and nonlinear uncertainties. A dedicated Matlab toolbox called IQClab was notably developed, enabling the H_2 performance of uncertain systems to be assessed [11]. But IQC approaches can be conservative. They can also be computationally costly, since the associated semi-definite programming problem grows quickly with the size of the uncertainty blocks and the number or states, in particular when high-order multipliers are needed.

Even if the aforementioned LMI-based approaches are efficient and accurate, they may not be the best approaches when it comes to being fast, while analyzing realistic systems characterized by high-dimensional models and multiple uncertainties. Alternative approaches, such as those proposed in [12], [2], make use of a combination of structured singular value (μ-) analysis and non-smooth optimization-based techniques. The latter are employed to find a function upper bounding the uncertain system, from which an upper bound on the worst-case H_2 norm is computed. Accurate results are generally obtained, but attention should be paid to systems with high peaks in their frequency response, e.g. with flexible dynamics, for which conservative results can be obtained.

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Moreover, one of the main drawbacks of these methods is that they do not provide guarantees on the computed bounds, because they depend on heuristics. Another alternative is proposed in [13], which relies on skew μ-analysis (also called ν-analysis) for computing an upper bound on the worst-case H_2 norm. Similar to [9], this algorithm relies on a branch-and-bound (B&B) scheme to reduce conservatism.

Most of the aforementioned analysis techniques aim at computing upper bounds on the worst-case H_2 norm of an uncertain linear system. Given this situation, and in the perspective of addressing the V&V challenge presented above, the main objectives of this paper are as follows:

- use existing characterizations of the upper bound on the worst-case H_2 norm to derive **satisfaction tests**, which guarantee that the H_2 norm of an uncertain system is higher than a certain performance level γ on a given uncertainty box, with as little conservatism as possible,
- extend these characterizations to derive **violation tests**, which guarantee that the H_2 norm of an uncertain system is lower than γ on a given uncertainty box, again with as little conservatism as possible,
- embed these satisfaction and violation tests in a **B&B scheme** to divide the entire uncertainty domain into smaller and smaller boxes, and finally compute tight bounds on the probability that the H_2 norm of an uncertain system exceeds γ.

Three different strategies are proposed. As detailed in Section IV, the first one uses [13] as a starting point, while the second and third ones build upon [12], [2]. Note that there are three decisive factors behind the choice to focus on μ-analysis based methods. Firstly, their computational efficiency. Secondly, the easy integration into a B&B scheme compatible with the probabilistic μ framework already deployed for stability, H_∞-performance and gain/phase/disk/delay margin analysis [14], [15]. And thirdly, the possibility of deriving a lower bound on the H_2 norm, which is guaranteed on a whole uncertainty domain, thanks to lemmas already used in prior probabilistic μ-studies [14].

The paper is organized as follows. The considered problem is formally stated in Section II. Section III gives an overview to the proposed solutions and highlights the contributions. The main theoretical results are then detailed in Section IV.

II. Problem statement

Let us consider the following continuous-time uncertain linear time-invariant (LTI) system:

$$\begin{cases} \dot{x} &= A(\delta)x + B(\delta)u \\ y &= C(\delta)x + D(\delta)u \end{cases} \tag{1}$$

The real uncertain parameters $\delta = (\delta_1, \ldots, \delta_N)$ are bounded and without loss of generality normalized, so that the whole set of admissible uncertainties is covered when $\delta \in \mathcal{B}_\delta = [-1\ 1]^N$. They are independent random variables, whose probability distributions f (typically, but not exclusively, uniform or truncated normal) are supported on the bounded interval $[-1\ 1]$. It is assumed that $A(\delta)$, $B(\delta)$, $C(\delta)$, $D(\delta)$ are polynomial or rational functions of the δ_i and that system (1) can be transformed into a Linear Fractional Representation (LFR) as shown in Fig. 1: the uncertainties are separated from the nominal LTI system $M(s)$ and isolated in a block-diagonal operator $\Delta = \text{diag}(\delta_1 I_{n_1}, \ldots, \delta_N I_{n_N}) \in \mathbb{R}^{p\times p}$, where I_{n_i} is the $n_i \times n_i$ identity matrix. The transfer between u and y is denoted $F_u(M(s), \Delta)$, where $F_u(.,.)$ is the upper linear fractional transformation. The set of matrices with the same block-diagonal structure as Δ is denoted $\mathbf{\Delta}$. Let $\mathcal{B}_{\mathbf{\Delta}} = \{\Delta \in \mathbf{\Delta} : \delta \in \mathcal{B}_\delta\}$ and $D_{\mathbf{\Delta}} = \{\Delta \in \mathbf{\Delta} : \delta \in D_\delta\}$ be the subsets of $\mathbf{\Delta}$ corresponding to $\mathcal{B}_\delta$ and to a given box $D_\delta \subset \mathcal{B}_\delta$.

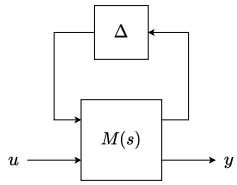

Fig. 1: Linear Fractional Representation

Definition 2.1: The frequency-limited H_2 norm of a continuous-time LTI system $G(s)$ is the restriction of its H_2 norm to a bounded frequency interval $\mathcal{I}_d = [\underline{\omega}\ \overline{\omega}]$, where $0 \leq \underline{\omega} < \overline{\omega} < +\infty$:

$$\|G(s)\|_{2,\mathcal{I}_d} = \left(\frac{1}{\pi} \int_{\underline{\omega}}^{\overline{\omega}} \text{tr}\left(G(\mathrm{j}\omega)^* G(\mathrm{j}\omega)\right) d\omega \right)^{\frac{1}{2}} \tag{2}$$

Using these definitions and notations, probabilistic finite-frequency H_2-performance analysis can be formalized as:

Problem 2.1: Given a performance level $\gamma > 0$ and a frequency interval $\mathcal{I}_d$, compute the probability $\overline{P}^{\gamma}_{\mathbf{\Delta},f}(M(s))$ that $\|F_u(M(s), \Delta)\|_{2,\mathcal{I}_d} > \gamma$ when $\Delta \in \mathcal{B}_{\mathbf{\Delta}}$.

A control system can then be rejected or validated depending on whether $\overline{P}^{\gamma}_{\mathbf{\Delta},f}(M(s))$ does or does not exceed a given predefined probability threshold ϵ. For instance, for spacecraft control, the desired γ-level, could be derived from the expected ratio between the power spectral density (PSD) of the pointing performance y and that of the input noise u. If this ratio should be satisfied for 99% of the cases among all uncertainty combinations, the value of ϵ is set to 1%.

Three different approaches are presented in Sections III-IV to compute tight bounds on $\overline{P}^{\gamma}_{\mathbf{\Delta},f}(M(s))$.

III. Overview of the proposed solutions

Over the past few years, substantial efforts have been made to implement probabilistic μ theory in an efficient toolbox, called the STOchastic Worst-case Analysis Toolbox (STOWAT) [16]. The first studies concerned stability and H_∞-performance [14], [17]. Gain, phase, disk and delay margins were considered next [15], [18], and the resulting algorithms were finally extended to the analysis of Multiple-Input Multiple-Output (MIMO) systems [19].

This paper aims to enrich this set of probabilistic μ-based tools by the introduction of various algorithms to analyze finite-frequency H_2-performance in a probabilistic way. These algorithms combine classical μ-based conditions

with a B&B scheme to explore the whole uncertainty domain $\mathcal{B}_\delta$. The H_2 norm is first computed at the center of $\mathcal{B}_\delta$, *i.e.* for the nominal system where $\Delta = 0$. This can be done e.g. with the Gramian-based method of [4]. If it is smaller (resp. larger) than γ, it is then checked whether the H_2-performance requirement is satisfied (resp. violated) on the entire domain $\mathcal{B}_\delta$ using sufficient conditions involving skew-μ upper bound computations, as detailed in Section IV. While performance satisfaction can be expressed fairly directly in the μ framework, this is not the case for performance violation, where the lowest finite-frequency H_2 norm over all possible uncertainty combinations in $\mathcal{B}_\delta$ is to be computed. Until now, no analysis technique was able to provide such a guaranteed lower bound, as most existing approaches focus on sampling-based techniques. Dedicated algorithms for this purpose are therefore one of the main contributions of this paper. If H_2-performance satisfaction or violation cannot be guaranteed, $\mathcal{B}_\delta$ is finally partitioned into smaller boxes, and this process is repeated until each box has guaranteed sufficient/insufficient H_2 norm or is small enough to be neglected (see Algorithm 1 in Section IV-E). Guaranteed upper and lower bounds on the *exact* probability of finite-frequency H_2-performance violation $\overline{P}^{\gamma}_{\mathbf{\Delta},f}(M(s))$ are finally obtained, based on the probability distributions f of the uncertain parameters δ, thus solving Problem 2.1.

IV. Probabilistic H_2-performance

Sections IV-B/C/D successively describe the three proposed approaches to check whether an H_2-performance requirement is satisfied or violated on an entire uncertainty box $D_\delta \subset \mathcal{B}_\delta$. Graphical representations of the underlying mechanisms are shown in Fig. 2. The three approaches have a common structure, which is first described in Section IV-A. All of them can finally be integrated into the B&B scheme of Algorithm 1, presented in Section IV-E, so as to cover the entire uncertainty domain $\mathcal{B}_\delta$.

A. Sufficient conditions for robust H_2 performance

Let $M(s) = \begin{bmatrix} M_{11}(s) & M_{12}(s) \\ M_{21}(s) & M_{22}(s) \end{bmatrix}$ be partitioned in accordance with Fig. 1. The following two assumptions hold for the rest of the study:

Assumption 4.1: The uncertain system (1) is Single-Input Single-Output (SISO), *i.e.* $u \in \mathbb{R}$ and $y \in \mathbb{R}$.

Assumption 4.2: The transfer function $M_{22}(s)$ is biproper, *i.e.* proper but not strictly proper.

Remark 4.1: Assumption 4.1 is required to use Proposition 3.1 of [14]. This central result in establishing a violation test for H_∞ performance is used again on several occasions in the sequel to obtain violation tests for H_2 performance. The main difference is that the LFR inversion it implies is only performed at one frequency for H_∞ performance, whereas it must be performed for all $\omega \in \mathcal{I}_d$ here. More precisely, it will be shown later that $F_u(M(s), \Delta)^{-1}$ must be computed. This requires $M_{22}(s)$ to be invertible, *i.e.* not to be strictly proper to avoid the use of descriptor systems, which justifies Assumption 4.2. Work is underway to relax these somewhat strong assumptions.

1) Satisfaction test: The three methods start with the definition of a frequency grid $I_{grid} \subset \mathcal{I}_d$. For each frequency $\omega_i \in I_{grid}$, with $i \in [1\ n_\omega]$, an upper bound $F_+(\omega_i)$ on the uncertain frequency response $F_u(M(\mathrm{j}\omega_i), \Delta)$ is computed:

$$F_+(\omega_i) \geq |F_u(M(\mathrm{j}\omega_i), \Delta)| \quad \forall \Delta \in D_{\mathbf{\Delta}} \tag{3}$$

A function $\overline{F_+} : \mathrm{j}\mathcal{I}_d \to \mathbb{C}$ is then derived such that:

$$|\overline{F_+}(\mathrm{j}\omega)| \geq |F_u(M(\mathrm{j}\omega), \Delta)| \quad \forall \Delta \in D_{\mathbf{\Delta}}, \forall \omega \in \mathcal{I}_d \tag{4}$$

A guaranteed upper bound $\overline{H_2^+}(\mathcal{I}_d)$ on the finite-frequency H_2 norm is finally computed by integrating $\overline{F_+}(\mathrm{j}\omega)$:

$$\begin{aligned} \overline{H_2^+}(\mathcal{I}_d) &= \left(\frac{1}{\pi} \int_{\underline{\omega}}^{\overline{\omega}} |\overline{F_+}(\mathrm{j}\omega)|^2 d\omega \right)^{\frac{1}{2}} \\ &\geq \underbrace{\left(\frac{1}{\pi} \int_{\underline{\omega}}^{\overline{\omega}} |F_u(M(\mathrm{j}\omega), \Delta)|^2 d\omega \right)^{\frac{1}{2}}}_{=\|F_u(M(s),\Delta)\|_{2,\mathcal{I}_d}} \quad \forall \Delta \in D_{\mathbf{\Delta}} \end{aligned} \tag{5}$$

The denser the grid I_{grid}, the more accurate the upper bound $\overline{H_2^+}(\mathcal{I}_d)$, but the longer the computational time. This leads to the following H_2-performance satisfaction test:

Proposition 4.1: The finite-frequency H_2 norm on $\mathcal{I}_d$ is lower than γ on a box $D_\delta \subset \mathcal{B}_\delta$, *i.e.* $\|F_u(M(s), \Delta)\|_{2,\mathcal{I}_d} \leq \gamma$ $\forall \Delta \in D_{\mathbf{\Delta}}$, if $\overline{H_2^+}(\mathcal{I}_d) \leq \gamma$.

2) Violation test: The approach to evaluate whether the H_2-performance requirement is violated on D_δ consists of the same steps as above. First, a lower bound on $F_u(M(\mathrm{j}\omega_i), \Delta)$ is computed for each $\omega_i \in I_{grid}$:

$$F_-(\omega_i) \leq |F_u(M(\mathrm{j}\omega_i), \Delta)| \quad \forall \Delta \in D_{\mathbf{\Delta}} \tag{6}$$

A function $\overline{F_-} : j\mathcal{I}_d \to \mathbb{C}$ is then derived such that:

$$|\overline{F_-}(\mathrm{j}\omega)| \leq |F_u(M(\mathrm{j}\omega), \Delta)| \quad \forall \Delta \in D_{\mathbf{\Delta}}, \forall \omega \in \mathcal{I}_d \tag{7}$$

A guaranteed lower bound $\overline{H_2^-}(\mathcal{I}_d)$ on the finite-frequency H_2 norm is finally obtained:

$$\begin{aligned} \overline{H_2^-}(\mathcal{I}_d) &= \left(\frac{1}{\pi} \int_{\underline{\omega}}^{\overline{\omega}} |\overline{F_-}(\mathrm{j}\omega)|^2 d\omega \right)^{\frac{1}{2}} \\ &\leq \underbrace{\left(\frac{1}{\pi} \int_{\underline{\omega}}^{\overline{\omega}} |F_u(M(\mathrm{j}\omega), \Delta)|^2 d\omega \right)^{\frac{1}{2}}}_{=\|F_u(M(s),\Delta)\|_{2,\mathcal{I}_d}} \quad \forall \Delta \in D_{\mathbf{\Delta}} \end{aligned} \tag{8}$$

This leads to the following H_2-performance violation test:

Proposition 4.2: The finite-frequency H_2 norm on $\mathcal{I}_d$ is larger than γ on a box $D_\delta \subset \mathcal{B}_\delta$, *i.e.* $\|F_u(M(s), \Delta)\|_{2,\mathcal{I}_d} \geq \gamma$ $\forall \Delta \in D_{\mathbf{\Delta}}$, if $\overline{H_2^-}(\mathcal{I}_d) \geq \gamma$.

As can be seen in Fig. 2, the definitions of $F_\pm(\omega_i)$ and $\overline{F_\pm}(\mathrm{j}\omega)$ differ for the three proposed approaches. They are detailed in Sections IV-B/C/D, along with other specific features of the various methods.

B. Method 1

1) Satisfaction test: The following value of $F_+(\omega_i)$ is clearly an upper bound on $|F_u(M(\mathrm{j}\omega_i), \Delta)|$ for all $\Delta \in D_{\mathbf{\Delta}}$:

$$F_+(\omega_i) = \max_{\Delta \in D_{\mathbf{\Delta}}} |F_u(M(\mathrm{j}\omega_i), \Delta)| \tag{9}$$

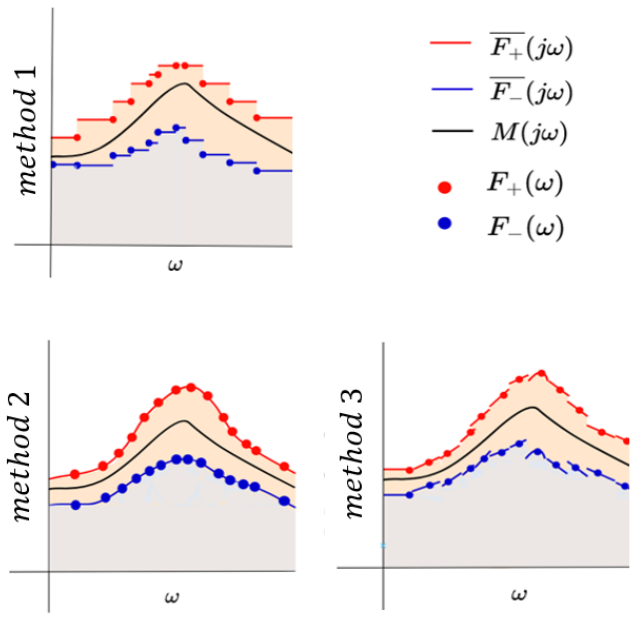

Fig. 2: Graphical representation of the bounds on the uncertain frequency response for the three proposed methods

Assuming that $M(s)$ is stable, the following equivalence based on the main loop theorem holds [20]:

$$\max_{\Delta \in D_{\mathbf{\Delta}}} |F_u(M(\mathrm{j}\omega_i), \Delta)| < \beta_i \;\Leftrightarrow\; \nu_{\mathbf{\Delta_a}}(M(\mathrm{j}\omega_i)) < \beta_i \quad (10)$$

where $\nu_{\mathbf{\Delta_a}}$ denotes the skewed structured singular value for the augmented uncertainty set $\mathbf{\Delta_a} = \mathrm{diag}(D_{\mathbf{\Delta}}, \mathbb{C})$. In short, $\nu_{\mathbf{\Delta_a}}(M(\mathrm{j}\omega_i))$ is the inverse of the size of the smallest complex uncertainty connected between y and u, which makes the interconnection of Fig. 1 unstable at frequency ω_i for some $\Delta \in D_{\mathbf{\Delta}}$. Its exact computation is NP hard, but efficient tools exist to compute a (usually tight) upper bound $\beta_i = \overline{\nu}_{\mathbf{\Delta_a}}(M(\mathrm{j}\omega_i))$ instead, see e.g. [21].

The most accurate way to define the function $\overline{F_+}(\mathrm{j}\omega)$ in this context would be to perform an infinite number of skew-μ upper bound computations to cover the entire $\mathcal{I}_d$. In practice, a piecewise constant function is used:

$$\overline{F_+}(\mathrm{j}\omega) = \sum_{i=1}^{n_\omega} \beta_i \mathbf{1}_{I^i_{valid}}(\omega) \quad (11)$$

where $\mathbf{1}_{I^i_{valid}}(\omega) = 1$ if $\omega \in I^i_{valid}$ and 0 otherwise. I^i_{valid} is a frequency interval containing ω_i on which β_i remains an upper bound on $\nu_{\mathbf{\Delta_a}}(M(\mathrm{j}\omega))$. It can be easily determined by analyzing the eigenvalues of a Hamiltonian-like matrix, as recalled in [21].

2) Violation test: For each frequency $\omega_i \in I_{grid}$, the following value of $F_-(\omega_i)$ is introduced:

$$F_-(\omega_i) = \min_{\Delta \in D_{\mathbf{\Delta}}} |F_u(M(\mathrm{j}\omega_i), \Delta)| \quad (12)$$

Equation (12) cannot be directly reformulated as a skew-μ problem, so the following lemma is introduced [20].

Lemma 4.1: If M_{22} is non-singular, then $F_u(M, \Delta)^{-1} = F_u(N, \Delta)$, where $N = \begin{bmatrix} M_{11} - M_{12}M_{22}^{-1}M_{21} & -M_{12}M_{22}^{-1} \\ M_{22}^{-1}M_{21} & M_{22}^{-1} \end{bmatrix}$.

Using Lemma 4.1, $F_-(\omega_i)$ can be reformulated as follows:

$$F_-(\omega_i) := \frac{1}{\max\limits_{\Delta \in B_\Delta} |F_u(N(\mathrm{j}\omega_i), \Delta)|} \quad (13)$$

and therefore be lower bounded by $\beta_i^{-1} = [\overline{\nu}_{\mathbf{\Delta_a}}(N(\mathrm{j}\omega_i))]^{-1}$ in the same way as above. Again, a piecewise constant function $\overline{F_-}(\mathrm{j}\omega)$ is defined:

$$\overline{F_-}(\mathrm{j}\omega) = \sum_{i=1}^{n_\omega} \frac{1}{\beta_i} \mathbf{1}_{I^i_{valid}}(\omega) \quad (14)$$

Remark 4.2: The transition from (12) to (13) is made possible by Assumption 4.1. If the uncertain system was MIMO, $\overline{\sigma}(F_u(M(\mathrm{j}\omega_i), \Delta))$ would appear instead of $|F_u(M(\mathrm{j}\omega_i), \Delta)|$ in (12), and therefore $\underline{\sigma}(F_u(N(\mathrm{j}\omega_i), \Delta))$ instead of $|F_u(N(\mathrm{j}\omega_i), \Delta)|$ in (13), where $\overline{\sigma}(.)$ and $\underline{\sigma}(.)$ denote the largest and smallest singular values. This would make the main loop theorem inapplicable. Then, $N(\mathrm{j}\omega_i)$ can be computed in (13) iff $M_{22}(\mathrm{j}\omega_i)$ is non-singular. In the SISO case considered here, it means that the nominal performance channel ($\Delta = 0$) should be nonzero, which is usually fulfilled in practice. However, a state-space representation of $N(s)$ is required to compute I^i_{valid} in (14), see e.g. [21]. This means that $M_{22}(s)$ should not be a strictly proper transfer function to avoid a descriptor form when performing the inversion, which is imposed by Assumption 4.2.

C. Method 2

1) Satisfaction test: $F_+(\omega_i)$ is now defined as follows:

$$F_+(\omega_i) = |M_0(\mathrm{j}\omega_i)| + \overline{\nu}_{\mathbf{\Delta_a}}\left(\tilde{M}(\mathrm{j}\omega_i)\right) \quad (15)$$

where $M_0(\mathrm{j}\omega)$ is the nominal frequency response and $F_u(\tilde{M}(\mathrm{j}\omega), \Delta)$ is the difference between the uncertain and the nominal frequency responses:

$$M_0(\mathrm{j}\omega) = F_u(M(\mathrm{j}\omega), 0) \quad (16)$$

$$F_u(\tilde{M}(\mathrm{j}\omega), \Delta) = F_u(M(\mathrm{j}\omega), \Delta) - M_0(\mathrm{j}\omega) \quad (17)$$

Lemma 4.2: $F_+(\omega_i) \geq |F_u(M(\mathrm{j}\omega_i), \Delta)| \;\; \forall \Delta \in D_{\mathbf{\Delta}}$.

Proof: Let $a \in \mathbb{R}_+$, $b \in \mathbb{C}$, $c \in \mathbb{C}$ and $d \in \mathbb{R}_+$. If $|b - c| \leq d$ and $a = |c| + d$, then $a \geq |b|$. This holds true since $a = |c| + d \geq |c| + |b - c| \geq |b|$. Applying this result with $a = F_+(\omega_i)$, $b = F_u(M(\mathrm{j}\omega_i), \Delta)$, $c = M_0(\mathrm{j}\omega_i)$, $d = \overline{\nu}_{\mathbf{\Delta_a}}\left(\tilde{M}(\mathrm{j}\omega_i)\right)$ proves that $F_+(\omega_i)$ meets equation (3). ■

With fitting, a function $\overline{F_+}(\mathrm{j}\omega)$ which satisfies (4) for all $\omega_i \in I_{grid}$ is obtained. To guarantee that $|\overline{F_+}(\mathrm{j}\omega)|$ is truly upper bounding $|F_u(M(\mathrm{j}\omega), \Delta)| \; \forall \omega \in \mathcal{I}_d$, an additional skew-$\mu$ analysis has to be performed, as stated in Proposition 4.3.

Proposition 4.3: $\overline{F_+}(\mathrm{j}\omega)$ satisfies equation (4) if:

$$\overline{\nu}_{\mathbf{\Delta_a}}\left(\begin{bmatrix} I_p & 0 \\ 0 & \overline{F_+}(\mathrm{j}\omega)^{-1} \end{bmatrix} M(\mathrm{j}\omega)\right) \leq 1 \quad \forall \omega \in \mathcal{I}_d \quad (18)$$

Proof: Equation (4) is equivalent to:

$$\max_{\Delta \in D_{\mathbf{\Delta}}} |F_u(M(\mathrm{j}\omega_i), \Delta)| \leq |\overline{F_+}(\mathrm{j}\omega)| \quad \forall \omega \in \mathcal{I}_d \quad (19)$$

According to the main loop theorem stated in equation (10), this is equivalent to:

$$\nu_{\mathbf{\Delta_a}}(M(\mathrm{j}\omega_i)) \leq |\overline{F_+}(\mathrm{j}\omega)| \quad \forall \omega \in \mathcal{I}_d \quad (20)$$

which is in turn equivalent to:

$$\nu_{\mathbf{\Delta_a}}\left(\begin{bmatrix} I_p & 0 \\ 0 & \overline{F_+}(\mathrm{j}\omega)^{-1}\end{bmatrix} M(\mathrm{j}\omega)\right) \leq 1 \quad \forall \omega \in \mathcal{I}_d \tag{21}$$

thanks to a classical property of the skewed structured singular value. ■

The Hamiltonian-based approach recalled in [21] is used again to ensure that the entire frequency range $\mathcal{I}_d$ is covered when checking (18). A sufficient condition to be able to do this is that $\overline{F_+}(s)$ is invertible, which is ensured by preventing it from being strictly proper during the fitting step. Then, if the skew-μ condition in Proposition 4.3 is not satisfied, the fit should be reconsidered. Minor modifications such as applying a constant multiplicative factor larger than 1 to slightly increase $|\overline{F_+}(\mathrm{j}\omega)|$ are often sufficient.

2) Violation test: A lower bound on $|F_u(M(\mathrm{j}\omega), \Delta)|$ for each $\omega_i \in I_{grid}$ is defined as:

$$F_-(\omega_i) = |M_0(\mathrm{j}\omega_i)| - \overline{\nu}_{\mathbf{\Delta_a}}\left(\tilde{M}(\mathrm{j}\omega_i)\right) \tag{22}$$

A lower bounding function $\overline{F_-}(\mathrm{j}\omega)$ is then fitted, which satisfies (7) for all $\omega_i \in I_{grid}$. Proposition 4.4 is finally applied to guarantee that $|\overline{F_-}(\mathrm{j}\omega)|$ is truly lower bounding $|F_u(M(\mathrm{j}\omega), \Delta)|$ $\forall \omega \in \mathcal{I}_d$.

Proposition 4.4: $\overline{F_-}(\mathrm{j}\omega)$ satisfies equation (7) if:

$$\overline{\nu}_{\mathbf{\Delta_a}}\left(\begin{bmatrix} I_p & 0 \\ 0 & \overline{F_-}(\mathrm{j}\omega)\end{bmatrix} N(\mathrm{j}\omega)\right) \leq 1 \quad \forall \omega \in \mathcal{I}_d \tag{23}$$

where $N(\mathrm{j}\omega)$ is obtained from Lemma 4.1.

As for Method 1, a state-space representation of $N(s)$ is required to apply the Hamiltonian-based approach and check the skew-μ condition of Proposition 4.4 on the whole frequency range (see Remark 4.2). It can be obtained if $M_{22}(s)$ is not strictly proper, which is guaranteed by Assumption 4.2.

D. Method 3

1) Satisfaction test: $F_+(\omega_i)$ is defined as follows:

$$F_+(\omega_i) = |M_0(\mathrm{j}\omega_i)| + \overline{\nu}_{\mathbf{\Delta_a}}\left(\tilde{M}(\mathrm{j}\omega_i)\right) = \alpha_i |M_0(\mathrm{j}\omega_i)| \tag{24}$$

where $\alpha_i = 1 + \dfrac{\overline{\nu}_{\mathbf{\Delta_a}}\left(\tilde{M}(\mathrm{j}\omega_i)\right)}{|M_0(\mathrm{j}\omega_i)|} > 1$. A function $\overline{F_+}(\mathrm{j}\omega)$ covering the entire frequency interval $\mathcal{I}_d$ is then defined as:

$$\overline{F_+}(\mathrm{j}\omega) = \overline{F_+^i}(\mathrm{j}\omega) \ \forall \omega \in I^i_{valid} \tag{25}$$

where:

$$\overline{F_+^i}(\mathrm{j}\omega) = \alpha_i M_0(\mathrm{j}\omega) \tag{26}$$

and I^i_{valid} is a frequency interval containing ω_i on which:

$$|\overline{F_+^i}(\mathrm{j}\omega)| \geq |F_u(M(\mathrm{j}\omega), \Delta)| \quad \forall \Delta \in D_{\mathbf{\Delta}} \quad \forall \omega \in I^i_{valid} \tag{27}$$

Inequality (27) can be checked by solving the skew-μ analysis problem of Proposition 4.5.

Proposition 4.5: Inequality (27) is satisfied if:

$$\overline{\nu}_{\mathbf{\Delta_a}}\left(\begin{bmatrix} I_p & 0 \\ 0 & M_0(\mathrm{j}\omega)^{-1}\end{bmatrix} M(\mathrm{j}\omega)\right) \leq \alpha_i \quad \forall \omega \in I^i_{valid} \tag{28}$$

The proof of Proposition 4.5 is similar to that of Proposition 4.3, with $\overline{F_+}(\mathrm{j}\omega)$ replaced with $\alpha_i M_0(\mathrm{j}\omega)$. I^i_{valid} can again be determined using the Hamiltonian-based approach recalled in [21]. A sufficient condition to be able to do this is that $M_0(s) = M_{22}(s)$ is invertible, which is ensured by Assumption 4.2.

2) Violation test: A lower bound on $|F_u(M(\mathrm{j}\omega_i), \Delta)|$ for each $\omega_i \in I_{grid}$ is defined as:

$$F_-(\omega_i) = |M_0(\mathrm{j}\omega_i)| - \overline{\nu}_{\mathbf{\Delta_a}}\left(\tilde{M}(\mathrm{j}\omega_i)\right) = \alpha_i |M_0(\mathrm{j}\omega_i)| \tag{29}$$

where $\alpha_i = 1 - \dfrac{\overline{\nu}_{\mathbf{\Delta_a}}\left(\tilde{M}(\mathrm{j}\omega_i)\right)}{|M_0(\mathrm{j}\omega_i)|} < 1$. Again, a discontinuous function $\overline{F_-}(\mathrm{j}\omega)$ covering the entire interval $\mathcal{I}_d$ is defined:

$$\overline{F_-}(\mathrm{j}\omega) = \overline{F_-^i}(\mathrm{j}\omega) \ \forall \omega \in I^i_{valid} \tag{30}$$

where:

$$\overline{F_-^i}(\mathrm{j}\omega) = \alpha_i M_0(\mathrm{j}\omega) \tag{31}$$

and I^i_{valid} is a frequency interval containing ω_i, on which:

$$|\overline{F_-^i}(\mathrm{j}\omega)| \leq |F_u(M(\mathrm{j}\omega), \Delta)| \quad \forall \Delta \in D_{\mathbf{\Delta}} \quad \forall \omega \in I^i_{valid} \tag{32}$$

Proposition 4.6: Inequality (32) is satisfied if:

$$\overline{\nu}_{\mathbf{\Delta_a}}\left(\begin{bmatrix} I_p & 0 \\ 0 & M_0(\mathrm{j}\omega)\end{bmatrix} N(\mathrm{j}\omega)\right) < \frac{1}{\alpha_i} \quad \forall \omega \in I^i_{valid} \tag{33}$$

As for Methods 1 and 2, a state-space representation of $N(s)$ is required to apply the Hamiltonian-based approach and compute the frequency interval I^i_{valid} on which the skew-μ condition (33) is valid. It can be obtained if $M_{22}(s)$ is not strictly proper, which is guaranteed by Assumption 4.2.

E. Overarching algorithm

For all three methods presented in Sections IV-B/C/D, the conditions for determining whether a given H_2 norm requirement is satisfied (Proposition 4.1) or violated (Proposition 4.2) on a box $D_\delta \subset \mathcal{B}_\delta$ are embedded into the high-level B&B scheme of Algorithm 1, which is similar to the one of [15]. $\mathcal{B}_\delta$ is divided into smaller and smaller boxes until each box has guaranteed H_2-performance or non-performance, or has a probability lower than a user-defined threshold p_{min}. But before using Algorithm 1, a preliminary stability analysis should be performed with Algorithm 1 of [14], leading to:

$$\mathcal{B}_\delta = D_s \cup D_{\overline{s}} \cup D_{s_u} \tag{34}$$

where D_s, $D_{\overline{s}}$ and D_{s_u} are the domains of guaranteed stability, guaranteed instability and undetermined stability respectively. Performance evaluation indeed only makes sense for stable systems, so the application of Algorithm 1 is restricted to D_s, leading to:

$$D_s = D_\gamma \cup D_{\overline{\gamma}} \cup D_{\gamma_u} \tag{35}$$

where D_γ, $D_{\overline{\gamma}}$ and D_{γ_u} are the domains of guaranteed H_2-performance satisfaction, guaranteed H_2-performance violation and undetermined H_2-performance, with corresponding probabilities $p(D_\gamma)$, $p(D_{\overline{\gamma}})$ and $p(D_{\gamma_u})$. These probabilities can be computed by integrating the probability density functions f of the uncertain parameters over the corresponding uncertainty domains. Combining (34) and (35), the final partitioning of $\mathcal{B}_\delta$ can be obtained as:

$$\mathcal{B}_\delta = D_\gamma \cup D_{\overline{\gamma}} \cup D_{\gamma_u} \cup D_{\overline{s}} \cup D_{s_u} \tag{36}$$

Guaranteed bounds on the exact probability $\overline{P}^{\gamma}_{\mathbf{\Delta},f}(M(s))$ of H_2-performance violation are finally obtained, which solves Problem 2.1:

$$p(D_{\overline{\gamma}}) \leq \overline{P}^{\gamma}_{\mathbf{\Delta},f}(M(s)) \leq p(D_{\overline{\gamma}}) + p(D_{\gamma_u}) + p(D_{s_u}) \quad (37)$$

Algorithm 1 Probabilistic finite-frequency H_2-performance

inputs: uncertain system $F_u(M(s), \Delta)$, probability distributions f, threshold γ

$\mathcal{L} \leftarrow \{\mathcal{B}_\delta\}$ ▷ *list of all boxes left to investigate*
$D_\gamma, D_{\overline{\gamma}}, D_{\gamma_u} \leftarrow \emptyset$
$p(D_\gamma), p(D_{\overline{\gamma}}), p(D_{\gamma_u}) \leftarrow 0$
while $\mathcal{L} \neq \emptyset$ **do**
  extract the box $D_\delta \subset \mathcal{L}$ with the highest probability
  compute the nominal H_2 norm at the center of D_δ
  if the nominal H_2 norm is $\leq \gamma$ **then**
    check H_2 norm satisfaction on D_δ with Proposition 4.1
    if guaranteed **then**
      add D_δ to D_γ and set $p(D_\gamma) \leftarrow p(D_\gamma) + p(D_\delta)$
    else
      declare current iteration as inconclusive
    end if
  else
    check H_2 norm violation on D_δ with Proposition 4.2
    if guaranteed **then**
      add D_δ to $D_{\overline{\gamma}}$ and set $p(D_{\overline{\gamma}}) \leftarrow p(D_{\overline{\gamma}}) + p(D_\delta)$
    else
      declare current iteration as inconclusive
    end if
  end if
  if current iteration is inconclusive **then**
    if $p(D_\delta) > p_{min}$ **then**
      select a direction for cutting D_δ (see e.g. [17])
      partition D_δ and add the resulting boxes into $\mathcal{L}$
    else
      add D_δ to D_{γ_u} and set $p(D_{\gamma_u}) \leftarrow p(D_{\gamma_u}) + p(D_\delta)$
    end if
  end if
end while

outputs: bounds on $\overline{P}^{\gamma}_{\mathbf{\Delta},f}(M(s))$ computed using (37)

V. Conclusion

Three novel algorithms have been developed to quantify the probability that a finite-frequency H_2 norm requirement is violated for an uncertain linear system. They all combine μ-analysis based computations to check whether the requirement is satisfied or violated on an uncertainty box, and a B&B scheme to explore the entire uncertainty domain. The key is the determination of accurate bounds on the frequency response of the uncertain system, which are guaranteed on the whole considered frequency range. The three proposed methods significantly differ in the way these bounds are defined. Preliminary Matlab code has been implemented with encouraging results, but it still needs to be optimized to minimize CPU time. The latter will nevertheless remain higher than for probabilistic H_∞ analysis. The entire considered interval must indeed be covered accurately, and not only the worst-case frequency, which requires a significant amount of skew-μ upper bound computations for each satisfaction or violation test. Finally, extensive work will be carried out to relax the assumptions considered in this paper, so as to make the proposed algorithms applicable to realistic systems.

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